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Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus*

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Abstract Chemicals are widely used in industries for various productions. The food chains are contaminated by Chemicals and toxic effects are displayed in animals at different levels of the chains including the human population. Cytochrome P450s are the single most important enzyme system. It is involved in the detoxification of xenobiotics. The main function of cytochrome P450 is the activation of molecular oxygen. Cytochrome P450 enzymes have the ability to react with a multitude of different chemical substrates. The P450 system contains a number of versatile isozymes in which they catalyze many types of reactions, including aliphatic and aromatic hydroxylations, epoxidations, N-oxidations, sulfoxidations, dealkylations, deaminations, dehalogenations, and others. Induction of hepatic cytochrome P450 enzymes in fish is also known to be sensitive biomarkers for monitoring polychlorinated biphenyl (PCB) or poly aromatic hydrocarbons. The present study was directed to analyse the fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus.* To study the EROD, MROD and BROD activity different concentrations (20, 40, 60, 80 and 100ppb) of fumaronitrile along with control group which was maintained for over 96 hrs. At every 1 hr. fishes were monitored with control and treatment groups.

Key Words Fumaronitrile, *Oreochromis mossambicus*, EROD, MROD, BROD

Introduction

Chemicals are widely used in Industries for production. The food chains are contaminated by Chemicals and toxic effects are displayed in animals at different levels of the chains including the human population. Chemicals biotransformation is the process by which xenobiotics are detoxified through enzymes to hydrophilic metabolites that are eliminated directly via renal elimination or after conjugation with endogenous cofactors.

Cytochrome P450s are the single most important enzyme system. It is involved in the detoxification of xenobiotics. P450 are so named because it is characterized by an intense absorption band at 450 nm when the protein is complex with carbon monoxide. The main function of cytochrome P450 is the activation of molecular oxygen. Cytochrome P450 enzymes have the ability to react with a multitude of different chemical substrates. This mechanism of reaction includes oxygen activation, splitting and formation of very reactive oxygen. The following thermodynamic equation demonstrates the catalytic power of cytochrome (Rein and Jung 1993):

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$$RH + O_2 + NADPH + H^+ \rightarrow ROH + H_2O + NADP^+$$

The P450 system contains a number of versatile isozymes in which they catalyze many types of reactions, including aliphatic and aromatic hydroxylations and epoxidations, N-oxidations, sulfoxidations, dealkylations, deaminations, dehalogenations, and others (Matsumura 1985). These isozymes are responsible for the oxidation of different substrates or xenobiotics and also responsible for different types of oxidative reactions of the same substrate.

Cytochrome P450s are typical monooxygenases. By using NADPH and molecular oxygen as co-substrates, cytochromes P450s catalyze oxygenation of various organic substrates. At the same time, these enzymes can utilize organic hydroperoxides and hydrogen peroxide as co-substrates in hydroxylation reactions. Cytochrome P450s, acting as an electron carrier can also reduce certain compounds. P450 dependent monooxygenations are accompanied by the release of superoxide radicals and hydrogen peroxide.

Cytochrome P450 enzymes are known under a variety of names, including monooxygenases, mixed function oxidases (MFOs), poly-substrate monooxygenases (PSMOs), microsomal oxidases, and heme thiolate proteins (Feyereisen 1999b). P450s are under complex and distinct control of endogenous signals, and various xenobiotic compounds have been shown to influence their activity. As an outcome, individual P450s can show diverse expression patterns related to life stages, sex, tissues, strains or diet (Scott and Wen 2001).

Because of their involvement in regulating the titers of endogenous compounds such as hormones, fatty acids and steroids, as well as in the catabolism of xenobiotics such as drugs, pesticides and plant toxins, these enzymes constitute an extremely important metabolic system (Scott 1999). P450 enzymes are found in the biosynthetic pathways of ecdysteroids and juvenile hormones, which are central to insect growth, development, and reproduction (Feyereisen 1999b). Cytochrome P450's represent an inducible enzyme system. Increased amounts of specific P450 isozymes are observed after treatment of organisms with a variety of environmental contaminants (Scott 1999).

Induction of cytochrome P450 by Chemicals and the consequences to toxicity are not well documented. However, there have not been much studies related to cytochrome P450 from different organisms involving exposure in some way to Chemicals. In the chironomid, Chironomus tentans, Fumaronitrile mediates the induction of certain cytochrome P450 enzymes which are responsible for oxidative activation of organophosphate insecticides into the more toxic O-analog metabolites (Pape-Lindstrom and Lydy 1997). Synergism of organophosphate and chlorinated hydrocarbon insecticides by both Fumaronitrile and other Chemicals has been found in Drosophila melanogaster (Lichtenstein *et al.*, 1979).

Kao *et al.*, (1995) opined that the induction of both cytochrome P450 activity and total P450 content is the consequence of incorporation of the herbicide Fumaronitrile by itself into the diet of the Southern armyworm (*Spodoptera eridania*). In rat livers, Fumaronitrile induced CYP1A2 and other enzymes as P-glycoprotein and glutathione-S-transferase demonstrates that Fumaronitrile is a specific inducer of some liver enzymes without any general toxic effect on the liver (Islam *et al.*, 2002).

The metabolism of xenobiotics is a two-phase process. The first phase involves reactions giving rise to more polar compounds. This may either result in reducing the effectiveness of the metabolised xenobiotic, which is described as bioelimination, or produce an active metabolite, in terms of both pharmacology and toxicology, which is described as bioactivation. The first-phase reactions include oxidation, reduction and hydrolysis and the greatest importance is ascribed to oxidation enzymes which are involved in the metabolism of the majority of xenobiotics.

Phase I enzymes and their isoforms metabolize pesticides and related compounds (Hodgson 2003). In second-phase reactions, the metabolites produced in the first phase or, occasionally, the original substances, are conjugated with products of the endogenous metabolism (glucuronate, glutathione, 3'-phosphoadenosine-5'-phosphosulphate, etc.) to give rise to polar compounds, subsequently eliminated from the body.

The cytochromes P450 are involved in a wide range of transformations of both endogenous and exogenous biological substances. Endoplasmic reticulum cytochromes are not substrate-specific. Each isoform participates in the metabolism of many different compounds. However, one substrate can also be metabolised by several different cytochrome isoforms. Almost every isoform has its specific substrate that can be used for its identification (Kvasniakova 1995; Anzenbacherova and Anzenbacher 1999, 2001; Lewis 2001). In contrast, bacterial and mitochondrial cytochrome P450 isoforms are highly substrate-specific (Lewis 2001).

Metabolism of xenobiotics is the primary function of isoforms (families CYP 1, CYP 2 and CYP 3). Isoforms are also involved in endogenous metabolic processes in substrates such as melatonin and estradiol (CYP 1A), testosterone (CYP 3A), catecholamines (CYP 2D), progesterone (CYP 2C, CYP 3A) and arachidonic acid (CYP 2E) (Lewis *et al.*, 2001).

Induction of hepatic cytochrome P450 enzymes in fish is also known to be sensitive biomarkers for monitoring polychlorinated biphenyl (PCB) or poly aromatic hydrocarbons (PAHs) (Ahokas *et al*., 1976). Cytochrome P450 not only used to monitors PCB but also Chemicals (Fumaronitrile ; terbuthylazine; ametryne; terbutryne; diuron), insecticides (carbaryl; fenthion), and fungicide (furametpyr) (Abass, *et al.*, 2007; Tang, *et al.*, 2002; Lang, *et al.*, 1997; Nagahori, *et al.*, 2000). The determination of ethoxy resorufin-O- deethylase (EROD) activity in fish has been used as an indicator of xenobiotics (Wassenberg et al., 2005). Methoxy resorufin-O-deethylase (MROD), penthoxy resorufin-O-deethylase (PROD), and bezyloxy resorufin-O- deethylase (BROD) activities are evaluated using PCB (Russell *et al.*, (2004).

EROD activity in fish is studied in field and laboratory conditions (Van der Oost *et al.*, 2003; Bucheli and Fent 1995; Engwall, *et al.*, 1997; Sundberg *et al.*, 2006). Brown trout (*Salmo trutta*) exposed to two moderately contaminated streams, shows differences in hepatic CYP1A protein and EROD activity between the streams (Honnen *et al.*, 2001; Behrens and Segner 2005). Atlantic salmon (*Salmo salar*), when exposed to tributyltin (TBT) leads to induction of BROD, MROD and EROD activity (Anne Skjetne Mortensen and Augustine Arukwe 2007).

Material and methods

Test Species

Oreochromis mossambicus

Tissue and microsomal fraction preparation

After 60, 80 and 100 days of exposure, *Oreochromis mossambicus* were dissected and liver, gonads and gill were taken out for microsomal fractions. Pooled tissues were homnogenised in 20 mM Tris–HCl (pH 7.6) (1g/4 ml) containing 0.25 M sucrose; 0.15 M KCl; 1 mM EDTA and 1 mM DTT and 100 lM PMSF. Microsomes were prepared by the procedure of Livingstone and Farrar (1984). All the preparation procedures were carried out at 4°C. The homogenate samples were centrifuged at first on 600 rpm for 10 min to sediment nuclei and cell fragments. Then, without transferring these fragments, samples were recentrifuged at 12,000 rpm for 45 min. The resultant pellet was resuspended in homogenizing buffer and recentrifuged at 12,000 rpm for 90 min, Resuspension buffer consisting of 20 mM Tris pH 7.6, 1 mM Dithiothreitol, 1 mM EDTA and 20% v/v glycerol were added into the remaining pellet. This resuspended pellet was used as a microsomal fraction.

Cytochrome isoform (CYP1A1, CYP1A2 and CYP2B6)

Ethoxy resorufin-O- dealkylase (EROD), Methoxy resorufin-O- dealkylase (MROD), and bezyloxy resorufin- O-dealkylase (BROD). All these enzymes involved in the elimination of xenobiotics. The formation of the product resorufin, due to cleavage of the substrate, was monitored at 575nm (the extinction coefficient of resorufin was 73 $mM^{-1} cm^{-1}$).

Ethoxy resorufin-O- dealkylation (EROD) activity was measured by the method of Koltz *et al.*, (1984), Methoxy resorufin-O-dealkylation (MROD), and Bezyloxy resorufin- O- dealkylation (BROD) activity was measured by the modified method of Lubet *et al.*, (1985). The reaction mixture consist of 50mM Tris, 25mM $MgCl_2$ buffer, 100µM NADPH, substrate (1µM 7- Ethoxy resorufin, 5µM 7- Methoxy resorufin and 5µM 7- Bezyloxy resorufin) were incubated for 3min at 37°C.

Statistical analysis

The values are expressed as mean ± SE. Differences between groups were assessed by one-way analysis of variance (ANOVA) using the Statistical Package for Social Sciences (SPSS) software package for windows (version 16.0). Post - hoc testing was performed for inter group comparisons using the least significant difference (LSD) test $p<0.05$ was considered statistically significant.

Result

Ethoxyresorifin O dealkylase

In the present study the specific activity of ethoxyresorifin O dealkylase (EROD) in the microsomal fraction of various tissues of *Oreochromis mossambicus* is analysed. Microsomal fraction exhibits higher activity in liver, gonads and gill (Fig1-3). There is a significant increase ($p< 0.05$) in EROD, activity in all the tissues of *Oreochromis mossambicus* exposed to Fumaronitrile when compared to control. The liver exerts higher activity when compared to gill and gonads. The significant increase ($p<0.05$) in enzyme activity is low in female compared with male *Oreochromis mossambicus.*

Methoxyresorufin O- dealkylase

Gradually increased activity of methoxyresorufin O- dealkylase (MROD) is observed in *Oreochromis mossambicus* during the experimental period. The specific activity of methoxyresorifin O deethylase in the microsomal fraction of various tissues like

liver, gonads and gill is shown in Fig (4-6). This enhanced activity in the liver is significantly higher ($p < 0.05$) when compared to gill and gonads.

Benzyloxyresorufin O- dealkylase

The specific activity of benzyloxyresorufin O- dealkylase (BROD) is analysed in the microsomal fraction of different tissues (liver, gonads and gill) were shown in Fig 7-9. When scrutinizing this enzyme level at different tissues the highest activity was measured in liver of *Oreochromis mossambicus.* EROD, MROD and BROD activity play a vital role in catalysing the conjugation and detoxification of toxic compounds. In this experiment the effect of Fumaronitrile on EROD, MROD and BROD activity of *Oreochromis mossambicus* was determined. In liver, gonads and gill of *Oreochromis mossambicus* treated with Fumaronitrile and control. CYP1A, CYP2A and CYP2B6 enzymes activities (EROD, MROD and BROD) are measured. A significant induction of EROD, MROD and BROD activity in *Oreochromis mossambicus* is exposed to Fumaronitrile.

Discussion

Most of the lipophilic chemicals like drugs, pesticides, carcinogens, environmental pollutants and naturally occurring compounds undergo enzyme-mediated oxidative, hydrolytic or conjugative biotransformation in liver and in extrahepatic tissues yielding more polar metabolites that can be easily excreted. Cytochrome P450 enzymes (CYPs) are active in the metabolism of wide variety of xenobiotics.

Induction of hepatic CYP1A1, CYP1A2 and CYP2B6 enzyme has been most commonly measured as EROD, MROD and BROD activity. CYP1A1 is reported as a biomarker for organic pollution in fish. Many researcher suggested the use of cytochrome P450 1A (CYP1A1) in fish as a biomarker for aquatic pollution (whyte, *et al.*, 2000; Arun and Subramanian 2007; Amutha *et al.*, 2009). However in some fish MROD and EROD activities show strong relations, suggesting the involvement of single CYP1A family isoenzymes (Smeets *et al.*, 2002).

In the present study activity of cytochrome P-450 isoforme is measurable in all the concentration of Fumaronitrile exposed *Oreochromis mossambicus*. During the treatment activity of MROD, EROD and BROD is gradually increased and it reaches a constant level until the end of the study period. Xenobiotics detoxification enzymes are analysed in various tissues of *Oreochromis mossambicus* such as liver, gonads and gill. All these enzyme activities were found to be higher in liver when compared to gonads and gill. When analysing the specific activity of ethoxyresorufin O- dealkylase, methoxyresorufin O- dealkylase, benzyloxyresorufin O- dealkylase in *Oreochromis mossambicus*, it is found that the activity is higher in the liver than that of other organs. This could be related to the fact that the liver is the major site for the production of biotransformation enzymes (Livingstone, 1991).

The liver is mainly responsible for the metabolism of toxic substances, including Fumaronitrile. Therefore the effects of Fumaronitrile on hepatic xenobiotics detoxification enzymes metabolism in *Oreochromis mossambicus* were screened. Such liver-detoxifying enzymes studies are very much important (Islam *et al.*, 2002). As a detoxification organ is necessary for the excretion of toxic substances in animals, the liver is probably the most common target

organ of toxicity in preclinical toxicology studies. Especially, the liver is major site of cytochrome P450 expression in teleost fish.

In this study, *Oreochromis mossambicus* is subjected to subchronic exposure of Fumaronitrile. The MROD, EROD and BROD levels in liver, gonads and gill of *Oreochromis mossambicus* is observed. The activity of MROD, EROD and BROD levels is increased in all concentration of Fumaronitrile exposed *Oreochromis mossambicus.* In the present study, *Oreochromis mossambicus* exposed to Fumaronitrile show sex-dependent induction of cytochrome P450-isoforms such as MROD, EROD and BROD activity. Males exposed to Fumaronitrile exhibited significantly reduced MROD, EROD and BROD activity while females exhibited significant induction of the same. Suppression of MROD, EROD and BROD activity in males is significant in all the concentration of Fumaronitrile in all the study periods. This result is supported by Le Goff *et al.,* (2006) who stated that sex dependence change in MROD, EROD and BROD activity. This result has been previously reported in many other fish species like Three-spined stickleback, Gasterosteus aculeatus and Zebrafish (Sanchez *et al.,* 2008; Beatrice Gagnaire *et al.,* 2013).

Increased liver weight has been associated with increased detoxification of Fumaronitrile . The increase in HSI following Fumaronitrile exposure is generally likely due to the enhanced activity of xenobiotics biotransformation (Zhou *et al.,* 1999; Sepulveda *et al.,* 2001). The increased activity of CYP1A1, CYP1A2 and CYP2B6 which is measured throughout experimental period may be due to accumulation Fumaronitrile Chemicals in the tissues of *Oreochromis mossambicus.* The exposure of Fumaronitrile is higher at the same time the activity of CYP1A1, CYP1A2 and CYP2B6 is also increased. This result clearly indicates that the activity of detoxification enzymes is concentration dependent, as was reported by Dong *et al.,* (2009) who stated that there is induction of cytochrome P450 isoenzymes in the microsomal fraction of zebrafish (*Danio rerio*) while exposed to Fumaronitrile. Mikula *et al.,* (2009) also observed similar induction of cytochrome P450 isoenzymes in common carp (*Cyprinus carpio*).

CYP1A1 is expressed at a very low level while CYP1A2 represents less than 10% of the total CYP enzymes (Pelkonen *et al*., 2008). The role of CYP1A1 and CYP1A2 has been shown to be important in the N-dealkylation of not only insecticides, but also Chemicals (Fumaronitrile) and fungicide (Lang *et al.,* 1997; Abass *et al*., 2009).CYP2B6 is previously reported as a minor hepatic CYP enzyme involved in detoxification of xenobiotics. It has been shown that CYP2B6 mediates the oxidative metabolism of Chemicals (Lang *et al.,* 1997).

The induction of CYP enzyme activities, particularly CYP1A1, in most vertebrate species is among the most biochemical response to organic contaminant exposure. The induction of MROD and EROD activities show strong relations in most fish, suggesting the involvement of a single CYP1A isoenzyme (Smeets *et al.,* 2002). Other studies have shown CYP1A2-like induction in several fish species (Leaver *et al.,* 1988; Celander and Forlin, 1991; Rotchell *et al.,* 2000). Moreover, catalytic activity of EROD, BROD and MROD showed minor effects in *Oreochromis mossambicus* exposed to Fumaronitrile . There is the decreased activity of BROD in *Oreochromis mossambicus* when compared to EROD and MROD.

Our results suggest that some important sex-related differences in xenobiotics metabolism (Fumaronitrile) occur in the *Oreochromis mossambicus*. Hence the present work concludes that many enzymes of cytochrome P-450 isoforme respond Fumaronitrile Chemicals, notable activity was found with MROD and EROD. It proves that MROD and EROD is a multi-substrate (not only the PCB and PAHs but also for Chemicals like Fumaronitrile) reactive enzyme. Therefore MROD can be considered as a potential biomarker for CYP mediated Fumaronitrile N- Dealkylation N-Deisopropylation in fish.

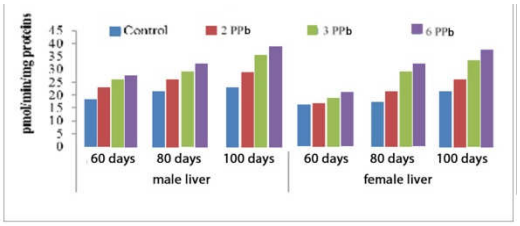

Fig. 1: EROD activity in liver of *Oreochromis mossambicus exposed to fumaronitrile*

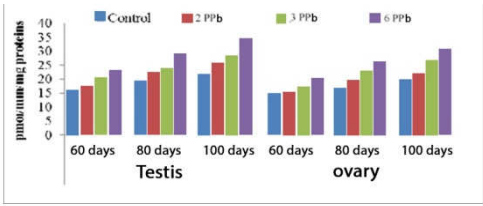

Fig. 2: EROD activity in gonads of *Oreochromis mossambicus exposed to fumaronitrile*

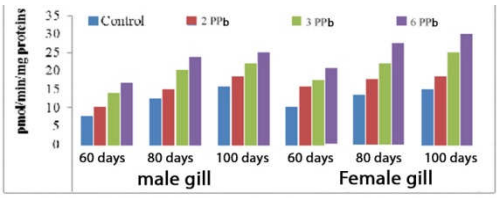

Fig. 3: EROD activity in liver of *Oreochromis mossambicus exposed to fumaronitrile*

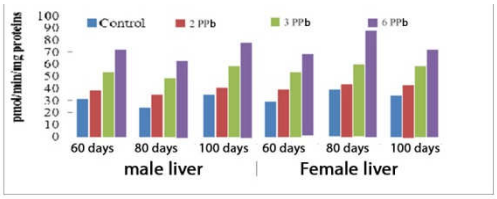

Fig. 4: MROD activity in liver of *Oreochromis mossambicus exposed to fumaronitrile*

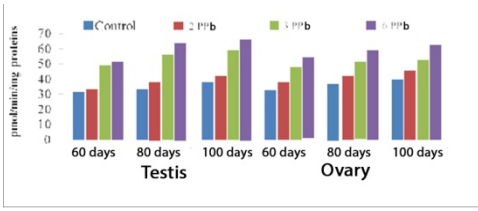

Fig. 5: MROD activity in gonads of *Oreochromis mossambicus exposed to fumaronitrile*

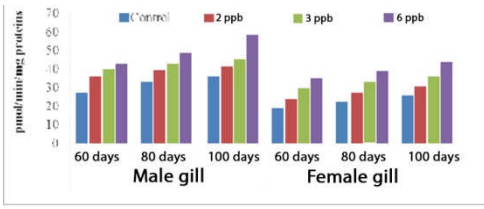

Fig. 6: MROD activity in gills of *Oreochromis mossambicus exposed to fumaronitrile*

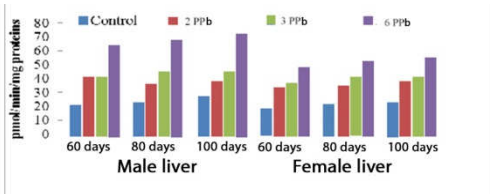

Fig. 7: BROD activity in liver of *Oreochromis mossambicus exposed to fumaronitrile*

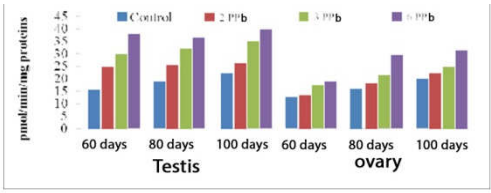

Fig. 8: BROD activity in gonads of *Oreochromis mossambicus exposed to fumaronitrile*

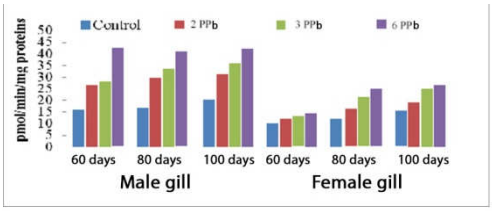

Fig. 9: BROD activity in gills of *Oreochromis mossambicus exposed to fumaronitrile*

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